

EDUGENIE – GOOGLE GEMINI POWERED LEARNING ASSISTANT

PHASE 2: Requirement Analysis Phase

1. Introduction

The requirement analysis phase identifies what EduGenie must provide to its users and what resources are required to develop the system. After finalizing the project idea, the team studied the needs of students and converted those needs into clear project requirements.

2. User Requirements

The main users are students and learners. They require a simple interface where they can enter questions and receive understandable educational assistance. The system should reduce unnecessary complexity and make interaction easy for users with different levels of technical knowledge.

3. Functional Requirements

EduGenie should accept user questions, process the input, communicate with the AI-powered system, receive a generated response, and display that response to the user. The system should maintain a clear flow between input and output.

4. Non-Functional Requirements

The system should be easy to use, reasonably responsive, readable, reliable, and maintainable. The interface should avoid unnecessary elements. The generated response should be displayed clearly so that students can understand it.

5. Hardware and Software Requirements

The project requires a suitable computer or mobile device for development and testing, internet connectivity, a development environment, and access to the required AI service. Supporting software and documentation tools are also needed.

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PHASE 2: Requirement Analysis Phase – CONTINUED

6. AI Requirements

Google Gemini is selected as the AI-powered component. The system requires a suitable method of sending user input to Gemini and receiving the generated response. Proper configuration and responsible use of AI-generated information are considered.

7. Input and Output Requirements

The main input is the student's question or learning request. The output is an AI-generated response. The system should handle normal questions clearly and should also provide sensible behaviour when the input is incomplete or unclear.

8. Scope and Limitations

The project focuses on demonstrating an AI learning assistant. It is not intended to replace teachers or guarantee that every AI-generated response is correct. Users should verify important information using reliable educational sources.

9. Requirement Summary

The analysis establishes a clear foundation for development: student-friendly interaction, question processing, Gemini-powered response generation, readable output, and basic reliability.

10. Conclusion

Requirement analysis converts the initial EduGenie idea into a practical set of expectations. These requirements guide the design, development, and testing phases and help the team avoid unnecessary features.